

SiyaBusa ERP

Enterprise Resource Planning System

AI Capabilities

✓ Production Ready

✓ 25+ Modules

✓ AWS Deployed

✓ 400+ APIs

✓ Multi-Tenant

✓ AI-Powered

Generated: January 2, 2026

Confidential & Proprietary

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AI ASSISTANT ENHANCEMENT

AI Assistant Enhancement - SiyaBusa ERP

"The Smartest Thing Ever" - Your First Point of Contact

Deployed: December 21, 2025

Status:  Live in Production

What Was Built

1. Comprehensive ERP Knowledge Base (`ERPKnowledgeBase.ts`)

The AI now knows EVERYTHING about the ERP system:

- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]

2. Machine Learning System (`AILearningService.ts`)

The AI learns and improves from every conversation:

- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]

3. Database Tables (PostgreSQL)

New tables created for AI learning:

```
ai_conversations      -- Stores all conversations for learning
ai_learned_patterns  -- Successful response patterns
ai_faq                -- Frequently asked questions
ai_user_preferences  -- Personalization per user
```

Pre-seeded with **15 common FAQs** about invoices, reconciliation, payroll, VAT, etc.

4. New API Endpoints

Endpoint	Method	Description
/api/ai/feedback/:id	POST	Record feedback (helpful/unhelpful)
/api/ai/faq	GET	Get FAQs for help center
/api/ai/analytics	GET	AI usage analytics
/api/ai/knowledge	GET	Full ERP knowledge base
/api/ai/knowledge/search	GET	Search knowledge base

How The AI Works Now

When User Asks a Question:

1. [object Object]
2. [object Object]
3. [object Object]
4. [object Object]
5. [object Object]
6. [object Object]

System Prompt Features:

The AI has been upgraded with:

- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]

Example Interactions

User: "How do I create an invoice?"

AI: "Go to Sales Hub → New Invoice. Select your customer, add line items with quantities and prices, review the totals including VAT, then click Save or Save & Send to email it directly. Would you like me to walk you through it step by step?"

User: "When is VAT due?"

AI: "VAT returns are due by the 25th of the month following your VAT period. For example, if your period ends on 30 November, your return is due by 25 December. Go to SARS Sentinel to see your upcoming deadlines and prepare your return."

User: "My trial balance doesn't balance"

AI: "Let's troubleshoot this. Check for: 1) Unposted journal entries, 2) Suspended transactions, 3) Opening balance errors, 4) Rounding issues. Go to Financial Hub → Reports → Trial Balance and look at any accounts with unusual balances. Would you like help identifying the discrepancy?"

Configuration Required

For the AI to work, ensure these environment variables are set:

```
OPENAI_API_KEY=sk-your-key-here
```

The AI uses GPT-4o-mini which costs approximately **R0.50-2 per customer per month** at typical usage.

Files Created/Modified

New Files:

- [object Object]
- [object Object]
- [object Object]

Modified Files:

- [object Object]
 - [object Object]
 - [object Object]
-

Analytics Available

Admins can see:

- [object Object]
- [object Object]
- [object Object]
- [object Object]
- [object Object]

Access via: `/api/ai/analytics`

Next Steps (Optional Enhancements)

1. [object Object]
 2. [object Object]
 3. [object Object]
 4. [object Object]
 5. [object Object]
-

Support

The AI is designed to be THE FIRST POINT OF CONTACT. Users should:

1. [object Object]
2. [object Object]
3. [object Object]
4. [object Object]

This reduces support load and provides instant, 24/7 assistance.

Live at: <https://primesources.site>

AI AGENTS QUICK START

AI Agents Module - Quick Start Guide

Installation (Complete)

The AI Agents module is already installed and integrated. All you need to do is configure your API keys.

1. Get API Keys

Option A: OpenAI (Recommended)

1. [object Object]
2. [object Object]
3. [object Object]

Option B: Anthropic (Alternative)

1. [object Object]
2. [object Object]
3. [object Object]

Option C: Both (Fallback support)

You can configure both. OpenAI will be used first, Anthropic as fallback.

2. Configure Environment

Add to your `.env` file:

```
# Required: At least one API key
OPENAI_API_KEY=sk-your-openai-key-here
# OR
ANTHROPIC_API_KEY=sk-ant-your-anthropic-key-here

# Optional: Configuration
AI_DEFAULT_MODEL=gpt-4
AI_MAX_TOKENS=2000
AI_TEMPERATURE=0.7
```

3. Deploy Database Schema

```
# SSH into your server
ssh ec2-user@51.21.219.35

# Navigate to backend
cd /home/ec2-user/backend

# Run migration
psql $DATABASE_URL -f database/migrations/018_ai_agents_module.sql

# Verify
psql $DATABASE_URL -c "SELECT agent_name FROM ai_agents;"
```

Expected output:

```
agent_name
-----
Sales Assistant
Procurement Assistant
Finance Assistant
HR Assistant
Compliance Assistant
Analytics Assistant
Treasury Assistant
Inventory Assistant
Logistics Assistant
(9 rows)
```

4. Restart Backend

```
pm2 restart aetheros-erp-backend
pm2 logs aetheros-erp-backend
```

Look for:

```
 Server running on port 3000
 ERP System initialized
```

5. Test AI Agents

Test 1: List Agents

```
curl -X GET http://51.21.219.35:3000/api/ai/agents \
-H "Authorization: Bearer YOUR_JWT_TOKEN"
```

Should return 9 agents.

Test 2: Chat with Sales Assistant

```
curl -X POST http://51.21.219.35:3000/api/ai/agents/SALES_ASSISTANT/chat \
-H "Authorization: Bearer YOUR_JWT_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "message": "What are the top 3 tips for closing deals?"
}'
```

Should return AI response with conversation ID.

Test 3: Get Suggestions

```
curl -X GET http://51.21.219.35:3000/api/ai/suggestions \
-H "Authorization: Bearer YOUR_JWT_TOKEN"
```

6. Frontend Integration

Add AI chat component to your frontend:

```
// Example: Chat with any agent
async function chatWithAgent(agentCode: string, message: string) {
  const response = await fetch(`/api/ai/agents/${agentCode}/chat`, {
    method: 'POST',
    headers: {
      'Authorization': `Bearer ${token}`,
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({ message })
  });

  return await response.json();
}

// Usage
const result = await chatWithAgent('SALES_ASSISTANT', 'Help me create a quotation');
console.log(result.response);
```

Available Agents

Agent Code	Module	Purpose
SALES_ASSISTANT	Sales	Quotations, forecasting, customer insights
PROCUREMENT_ASSISTANT	Purchase	PO optimization, supplier analysis
FINANCE_ASSISTANT	Financial	Reconciliation, variance analysis
HR_ASSISTANT	HR	Onboarding, payroll, policies
COMPLIANCE_ASSISTANT	Compliance	Risk assessment, deadlines
ANALYTICS_ASSISTANT	Reports	Data insights, trends
TREASURY_ASSISTANT	Treasury	Cash flow, FX, investments
INVENTORY_ASSISTANT	Inventory	Stock optimization, demand forecasting
LOGISTICS_ASSISTANT	Logistics	Route optimization, fleet tracking

Common Use Cases

Sales: Generate Quotation

```
POST /api/ai/sales/generate-quotation
{
  "customerId": "cust-123",
  "items": [{"productId": "prod-456", "quantity": 10}]
}
```

Compliance: Check Status

```
GET /api/ai/compliance/check
```

Finance: Explain Variance

```
GET /api/ai/finance/explain-variance?accountCode=4000&period=2024-01
```

Sales: Forecast

```
GET /api/ai/sales/forecast?period=quarter
```

Monitoring

Check Usage

```
SELECT
  a.agent_name,
  SUM(ua.total_messages) as messages,
  SUM(ua.total_tokens) as tokens
FROM ai_usage_analytics ua
JOIN ai_agents a ON ua.agent_id = a.agent_id
WHERE ua.usage_date >= CURRENT_DATE - INTERVAL '7 days'
GROUP BY a.agent_name;
```

View Recent Conversations

```
SELECT
  a.agent_name,
  c.conversation_title,
  c.last_message_at,
  (SELECT COUNT(*) FROM ai_messages WHERE conversation_id = c.conversation_id) as message_count
FROM ai_conversations c
JOIN ai_agents a ON c.agent_id = a.agent_id
ORDER BY c.last_message_at DESC
LIMIT 10;
```

Troubleshooting

Error: "No AI provider configured"

Solution: Add `OPENAI_API_KEY` or `ANTHROPIC_API_KEY` to `.env` and restart backend

Error: "Agent not found"

Solution:

```
# Check if agents exist
psql $DATABASE_URL -c "SELECT * FROM ai_agents WHERE is_active = true;"

# If empty, re-run migration
psql $DATABASE_URL -f database/migrations/018_ai_agents_module.sql
```

Slow responses

Solution:

- [object Object]
- [object Object]

High costs

Solution:

- [object Object]
- [object Object]
- [object Object]

Next Steps

1. [object Object]
2. [object Object]
3. [object Object]
4. [object Object]

Cost Estimation

OpenAI GPT-4 Pricing (as of Nov 2024):

- [object Object]
- [object Object]

Example: 100 conversations/day with 500 tokens each = \$3/day = \$90/month

Optimization:

- [object Object]
- [object Object]
- [object Object]

Support

Full documentation: </backend/AI-AGENTS-DOCUMENTATION.md>

Status:  READY FOR DEPLOYMENT

Version: 1.0.0

Date: November 2024